

DEBT, GROWTH, AND INEQUALITY

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INTRODUCTION

- In the aftermath of the Great Recession and the COVID-19 pandemic public debt levels have soared to unprecedented levels. The extraordinary high debt levels and seeming fragility of the global economy has brought public and academic debate to reconsider the long-run implications of fiscal spending. Modern arguments can be traced back to Buchanan (1958), Meade (1958), and Modigliani (1961), although the issue goes back as far as Adam Smith.
- The empirical relationship between public debt and economic growth in OECD and developing countries has been studied exhaustively.
- What has not been as extensively covered is the relationship between gross government debt, economic growth, and the distribution of income.

- We construct a rich dataset combining data on gross government debt from the International Monetary Fund International Financial Statistics database, growth data from the World Development Indicators database maintained by the World Bank, and additional data from the Penn World Tables (10.0).
- For our response variable, we utilize the data provided by the Standardized World Income Inequality Database (Solt 2020), which is a comprehensive database for country-wide estimates of the Gini coefficient for 196 countries covering 1960 to the present.
- Since our analysis allows for slope heterogeneity across countries, we need a sufficient number of time periods to estimate country-specific coefficients. To this end, we include only countries in our sample for which we have at least 30 consecutive annual observations on debt, the Gini coefficient, and GDP.

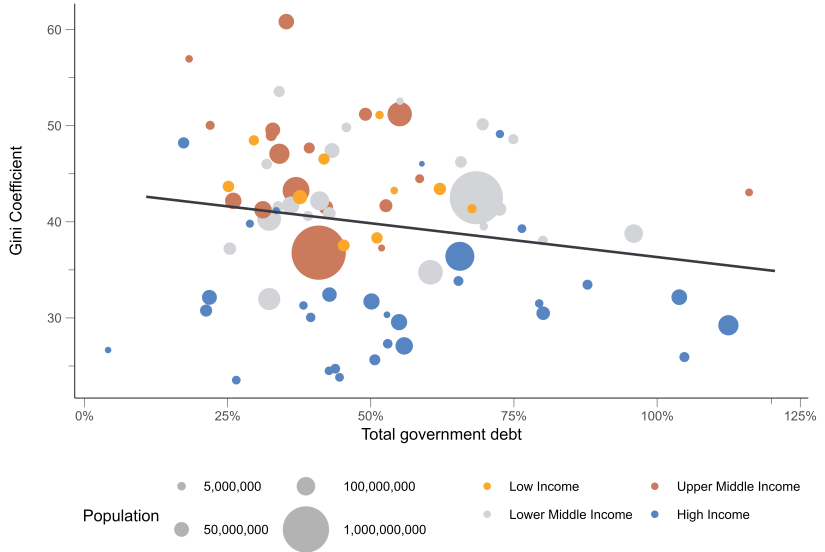
Table 1: Included Countries

Region	Upper Middle Income	High Income	Low Income	Lower Middle Income	Total
Latin America & Caribbean	11	5	0	2	18
East Asia & Pacific	4	5	0	1	10
Europe & Central Asia	1	18	0	0	19
Sub-Saharan Africa	3	0	10	9	22
South Asia	0	0	0	5	5
North America	0	2	0	0	2
Middle East & North Africa	0	0	0	6	6
Total	19	30	10	23	82

Table 2: Income Classification of Countries

High income	Low income	Lower middle income	Upper middle income
Australia	Burkina Faso	Bangladesh	Argentina
Austria	Central African Republic	Bolivia	Brazil
Belgium	Gambia	Cote d'Ivoire	Botswana
Barbados	Madagascar	Algeria	China
Canada	Mali	Egypt	Colombia
Switzerland	Malawi	Ghana	Costa Rica
Chile	Niger	Honduras	Ecuador
Cyprus	Rwanda	India	Guatemala
Germany	Sierra Leone	Iran	Indonesia
Denmark	Uganda	Jordan	Jamaica
Spain		Kenya	Mexico
Finland		Sri Lanka	Mauritius
France		Lesotho	Malaysia
United Kingdom		Morocco	Peru
Greece		Mauritania	Paraguay
Ireland		Nigeria	El Salvador
Italy		Nepal	Thailand
Japan		Pakistan	Turkey
South Korea		Philippines	South Africa
Luxembourg		Senegal	
Netherlands		Tunisia	
Norway		Zambia	
New Zealand		Zimbabwe	
Panama			
Portugal			
Singapore			
Sweden			
Trinidad and Tobago			
Uruguay			
United States			

Figure 1: Debt and Inequality



- Our sample restrictions provide us a dataset of 82 countries, with a time span of 1973 to 2019.
- We first consider the long-run effects of debt and output growth on the distribution of income using a traditional panel fixed-effects approach, in the following equation:

$$y_{it} = \alpha_i + \lambda_t + \gamma d_{it} + \beta \mathbf{x}_{it} + u_{it} \quad (1)$$

where y_{it} is the log of the Gini coefficient and $\mathbf{x}_{it}$ a vector of exogenous variables including the the log of debt to GDP ratio d_{it} , and g_{it} the growth rate. α_i and λ_t are country and time specific fixed-effects.

RESULTS

- As a baseline, we fit a pooled model of all 82 countries together. Then we fit separate models based on the IMF income classifications.
- We also consider the interaction effect between the debt level and the growth rate.
- By fitting these models to different income groups, we aim to uncover distinct patterns and relationships between income inequality and economic indicators across varying levels of economic development.

Table 3: Regression Results

	(1)	(2)	(3)	(4)	(5)
Total debt	-0.004 (0.045)	0.002 (0.081)	0.043** (0.025)	0.010 (0.031)	0.063** (0.031)
GDP	-0.307* (0.289)	-0.307 (0.342)	0.038 (0.162)	0.200** (0.075)	-0.055 (0.216)
Human Capital	-0.052 (0.130)	-0.319** (0.017)	-0.168*** (0.152)	0.167** (0.062)	-0.078 (0.101)
Debt $\times$ GDP	0.226 (0.379)	0.279 (0.480)	-0.147* (0.362)	-0.263** (0.111)	0.178 (0.398)
Observations	3,276	760	1,199	397	920
R^2	0.837	0.795	0.938	0.898	0.923
Adjusted R^2	0.831	0.777	0.934	0.882	0.917

Note: Column (1) is the full dataset, (2) are Upper Middle Income, (3) High Income, (4) Low Income, and (5) Lower Middle Income countries.

IMPLICATIONS

- For the overall model including all countries, debt, GDP, and human capital did not significantly affect income inequality.
- However, in upper-middle and high-income countries, higher human capital significantly reduces inequality, while in low-income countries, both GDP and human capital increase inequality.
- Additionally, for low-income countries, higher debt mitigates the impact of GDP on increasing inequality. Lower-middle-income countries showed a marginally significant positive effect of debt on inequality.

CONCLUSION

- Investments in education and skills (human capital) are effective in reducing income inequality in upper-middle and high-income countries but can increase inequality in low-income contexts due to unequal access.
- Economic growth alone, represented by GDP, can exacerbate inequality in low-income countries without inclusive growth policies.
- Public debt generally does not directly affect inequality, but responsible debt management can help mitigate the adverse effects of economic growth on income distribution, particularly in low-income countries.
- Policies tailored to the specific needs of different income groups are essential for effectively addressing income inequality.